

Pre-Thesis I Report

Deep Learning Techniques for Multi-Modal Analysis in Cardiovascular Risk Assessment

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Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Abstract

Many people die every year from Cardiovascular Disease (CVD) which is a major cause of death worldwide, highlighting the critical need for reliable, accurate and rapid risk monitoring methodologies. The existing methods for diagnosing CVDs mostly depend on the single modal models which limit their performance in case of classifying different types of CVDs systematically. The study mainly inquires about multi-modal analysis using deep learning models to categorize the CVDs distinctively, increasing the accuracy and stability rate. Multiple advanced Deep learning computational models or frameworks including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks and Transformer-based models extract vital data patterns from multiple information sources. The study investigates the primary obstacles which consist of heterogeneous data, incomplete patient information and unclear model understanding for deep learning applications in medical scenarios. Performance is being evaluated based on classification accuracy, F1 score, precision and recall through benchmark datasets assessment. The proposed work advances CVD diagnostic capabilities through specific disease identification with data-driven personal treatment selection as it provides healthcare providers with an operational solution at scale.

Keywords: Cardiovascular disease, Deep learning, Multi-Modal, Classification, Electrocardiogram, Magnetic resonance imaging, Convolutional neural network

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Nomenclature

The next list describes several symbols & abbreviation that will be later used within the body of the document

ACO Ant Colony Optimisation

ANN Artificial Neural Network

ASCVD Atherosclerotic Cardiovascular Disease

AUROC Area Under The Receiver Operating Characteristics Curve

CNN Convolutional Neural Network

CRF Clinical Risk Factors

CVD Cardiovascular Disease

DNN Deep Neural Network

ECG Electrocardiogram

EHR Electronic Health Record

FP Fundus Photograph

LSTM Long Short-Term Memory

MLP Multilayer Perceptron

MRI Magnetic Resonance Imaging

PCE Pooled Cohort Equation

PCG Phonocardiogram

RNN Recurrent Neural Network

SHAP SHapley Additive exPlanations

XAI Explainable AI

Chapter 1

Introduction

1.1 Background

Cardiovascular disease refers to a range of disorders affecting the heart and blood vessels. Cardiovascular disease is a leading cause of death worldwide. According to the World Health Organization, CVD causes 31% of deaths annually in the world [6]. Moreover, many people remain undiagnosed for a long time unless any life-threatening event happens. Also, those who survive heart attack or stroke often suffer from long-term life-threatening complications resulting in a reduced lifetime. CVD also economically impacts people, with billions spent annually on treatment, hospitalizations, and long-term care. Despite significant advancements in medical science, CVD remains a major concern due to its complex nature and the challenges in early detection. Several models have been developed for early diagnosis of cardiovascular disease. However, the models give a prediction of whether the person is suffering from any cardiovascular disease or not. The models do not clearly classify cardiovascular diseases into different types. We are developing a technique to detect different types of CVD. For example, arrhythmia, cardiomyopathy, coronary artery disease, rheumatic heart disease etc.

1.2 Rational of the Study or Motivation

Accurately identifying and categorizing different types of cardiovascular disease (CVD) is essential for early diagnosis and personalized treatment plans. Different types of CVD have unique causes, risk factors, and treatment approaches. For instance, when arteries become clogged with plaque, it causes coronary artery disease (CAD). To restore blood flow in the heart, angioplasty might be required. But rheumatic heart disease is caused by rheumatic fever which can ultimately result in heart failure. It needs different treatment. Predicting the specific type of CVD enables doctors to choose appropriate treatment plans and reduces the need for unnecessary procedures. Correct classification enables timely and appropriate intervention, reducing life-threatening complications.

1.3 Problem Statement

Nearly 17.3 million people die from Cardiovascular diseases (CVDs) each year globally [6]. Even with advances in the medical sector, quickly identifying and accurately assessing the risks of cardiovascular diseases (CVDs) is still a big challenge. The case of single-method analysis from electrocardiograms (ECG) together with echocardiography and clinical biomarkers results in diagnostic gaps and diagnostic errors. The reliance on handcrafted feature extraction methods in conventional machine learning models impairs their performance in identifying and monitoring complex relationships between physiological patterns during CVD prediction.

Previous research tried to use CNNs, LSTMs, and ensemble models to use deep learning for cardiovascular risk prediction. The current approaches to medical decision-making encounter major weaknesses which encompass relying on a single data type while operating on biased data. The systems also lack the ability to explain their decision processes. The promise of multi-modal frameworks that combine ECG signals with clinical demographics and imaging data has proven limited by challenges with optimal feature fusion together with real-time adaptability and interpretability needs for clinical deployment. The current models face difficulties in performing generalizations across different patient groups because they cannot handle heterogeneous datasets and unbalanced data distribution patterns.

To fill these critical gaps, this research presents a deep learning-based multi-modal analysis framework that leverages ECG signals, medical imaging and clinical biomarkers to improve cardiovascular risk assessment [1]. The study attempts to develop advanced feature fusion techniques, transformers, attention mechanisms, and also graph-based neural networks to improve predictive accuracy and interpretability. The proposed model uses state-of-the-art Deep learning architectures for a better reliable, more explainable and more scalable solution for cardiovascular disease prediction to contribute to early intervention and personalized healthcare.

1.4 Objective

Medical diagnostics have benefited from deep learning advancements yet cardiovascular disease risk analysis continues to be complex because of the numerous types of medical data. This research creates and deploys a sophisticated Deep learning structure that performs multi-modal assessment of cardiovascular risk through ECG data integration with medical images and clinical biomarkers [10]. The proposed framework will use transformer-based architectures, GNNs and attention mechanisms to achieve high predictive accuracy together with clinical interpretation capabilities. The proposed system will function across different computational scales with an efficiency that permits its deployment within remote and clinical healthcare environments. The model performs multi-classification of cardiovascular conditions along with risk severity assessment through patient-specific data inputs to support medical decision-making about early accurate interventions. Along with explainable AI techniques, the research will use transparent methods for predictions which can be interpreted clinically. The proposed research will deal with ethical concerns about data privacy as well as fairness in diagnosis through AI to establish trust in

automated CVD risk assessment systems. The other key goals of this study are outlined below:

1. Study the success rates of deep learning models which include transformers, CNNs and hybrid architectural designs for CVD risk evaluation that integrates multiple data types.
2. A deep learning framework should merge ECG data with medical images and clinical biomarkers to assess complete cardiovascular hazards [10].
3. The use of sophisticated feature fusion methods featuring attention systems and graph-based models should be developed for optimal predictive accuracy together with robust performance [5].
4. The model should demonstrate performance evaluation against prevailing state-of-the-art approaches in all three domains of accuracy, interpretability and computational speed.
5. Search for model limitations then enhance the capabilities to function properly throughout patient groups with different backgrounds using various medical information systems.
6. The system requires a friendly interface to allow real-time interactions and visual analytics for medical professionals.

1.5 Methodology in Brief

The study uses Deep learning methods through multi-modal analysis to integrate various patient information sources for enhancing cardiovascular risk evaluation. The study develops its research using experimental methods alongside data-driven procedures where state-of-the-art Deep learning frameworks improve both diagnostic precision and interoperability.

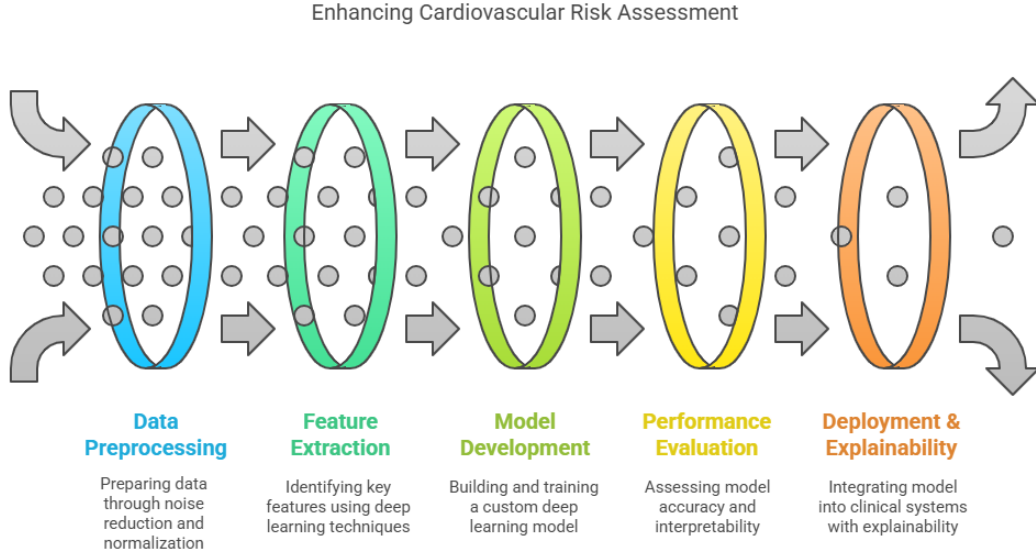


Figure 1.1: Methodology Workflow

The above shown workflow is described below:

- 1. Data Collection & Preprocessing:** The data collection consists of ECG signals combined with medical imaging information and clinical biomarkers for training model evaluation. The dataset needs preprocessing to reduce noise while normalizing data before applying augmentation techniques which improve model performance consistency.
- 2. Feature Extraction & Selection:** An advanced deep learning method will extract vital characteristics from both ECG waveforms and imaging scans as well as clinical parameters during the feature engineering process. The technique of dimensionality reduction will select the most important features which will lead to accurate predictions.
- 3. Model Development & Training:** A tailored Deep learning model will unite CNNs along with Transformers and Graph Neural Networks (GNNs) for performing imaging analysis together with sequence-based ECG analysis and relational data processing. Adaptive learning rate optimization together with regularization techniques will be implemented during training to stop the model from becoming overfitting.

4. **Performance Evaluation:** Cross-validation approaches will validate the model before evaluating its performance through accuracy together with precision and recall, F1-score and AUC-ROC metrics. Statistical evaluation of traditional ML techniques against this model will exhibit the new predictive strength.
5. **Deployment & Explainability:** The framework will be implemented through a user-friendly system which offers real-time interpretation capabilities powered by SHAP integrated with attention techniques. The integration goes to help healthcare providers to better understand and trust in AI-powered forecasts.

1.6 Scopes and Challenges

The research develops a Deep learning multi-modal framework for cardiovascular risk assessment through the unification of ECG signals and medical images with clinical biomarkers. The model implements AI transformers and CNNs together with GNNs to improve the early detection of CVDs as well as the diagnosis and risk assessment. The system can provide scalability because it applies to clinical environments as well as telemedicine and wearable health tracking devices which make it easy for healthcare providers to understand. Key challenges and constraints are:

1. Organizing and aligning various data sources to achieve successful feature integration.
2. Resource requirements for training deep learning multi-modal models remain high because of their computational complexity.
3. The model's robustness faces possible degradation from limited dataset diversity that can cause performance bias.
4. Ensuring that AI predictions are easy to understand and can be functioned upon in the medical sector.
5. Maintaining data security and privacy regulations (HIPAA, GDPR).

Chapter 2

Literature Review

2.1 Preliminaries

Electrocardiogram (ECG) signals are a heart's electrical activity changing over time and a record of information about the heart rhythm, conduction abnormality, and general cardiac health. Medical diagnostics provided by them are employed to identify conditions such as arrhythmias, myocardial infarction and other cardiovascular diseases.

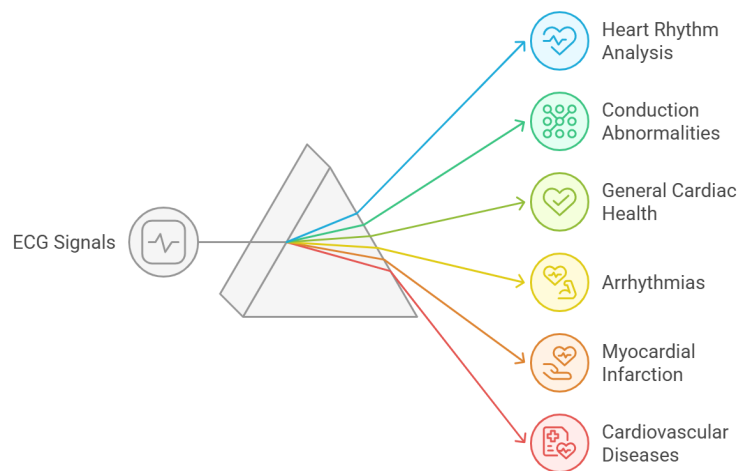


Figure 2.1: ECG Signals

Phonocardiogram(PCG) signals are recorded as graphical displays by placing a microphone on the chest wall. The heart sound signals with their acoustic events such as the "lub" (S1) and "dub" (S2) sounds along with any abnormal heart noises get captured through these signals. The assessment and diagnosis of heart function, as well as the diagnosis of heart diseases and heart health monitoring, rely on PCG signals in the field of cardiology. These signals enable experts to identify irregular heart rhythms together with problems associated with heart valves.

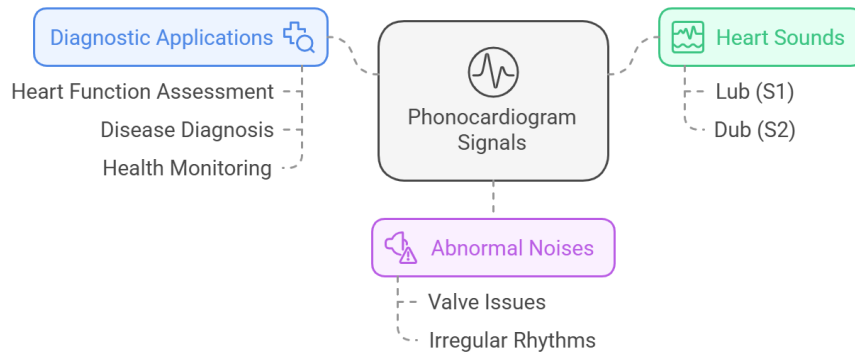


Figure 2.2: PCG Signals

Convolutional Neural Network (CNN) is a deep learning model built to analyze structured grid data, such as images. It is particularly good at identifying patterns, detecting different objects and classifying images into different groups by learning spatial hierarchies of features.

Graph Neural Networks (GNNs) are the kind of deep learning models processing and analyzing graph structured data where interconnections between a set of nodes are modelled. They employ message passing schemes for aggregating and updating node features which enable them for social network analysis, molecular property prediction, and recommendation system types of applications.

Deep Neural Network (DNN) is an advanced neural network. It has multiple hidden layers due to which it can learn complex patterns from data. It works by passing information through multiple layers of connected nodes, where each layer pulls out meaningful features step by step. The deeper the network, the better it can capture complex relationships, making DNNs powerful in recognizing and classifying images, processing speech and predicting diseases.

SHapley Additive exPlanations (SHAP) is a method for predicting interpretations for machine learning models. It is derived from game theory and allows explanation of individual prediction using the output of a model from the input feature. The SHAP values mainly indicate the relevant feature that contributes to the model prediction.

2.2 Review of Existing Research

The study [1] mainly focuses on the significance of detecting CVD in the early stage using CraftNet, a deep learning ensemble model as the existing models struggle with dissimilar arrangement of classes and categories as well as with the new, unseen data which are outside the training data while analyzing ECG signals. This study used the MIT-BIH dataset containing 48 30-minute-long ECG records of 47 patients which is at 360HZ. The heartbeat classes are expressed in 4 categories which include N(normal beat), S(supraventricular ectopic beat), V(ventricular ectopic beat), F(fusion beat) and Unknown beat. The dataset contains 90% N-type heartbeats comprising data imbalance. To alleviate the problem of data imbalance and for increasing accuracy rate CraftNet merged both handcrafted features such as -R-R intervals, wavelet transformations, higher order statistics and morphological data with DDAG structure and novel P-S loss(pseudo-SVM loss). The model had shown average sensitivity of 89.25% and specificity of 95.79% which is better than the established models establishing robustness against imbalanced data. Despite showing remarkable results the main drawback of this model is it is not completely automated as it uses handcrafted features which is hindering its adaptability to new datasets and not all datasets are prone towards manual feature extraction. Moreover, it has a complex training process for using mixed loss functions requiring more time in careful fine tuning though it is improving the performance. Not only this there may be high possibility of not handling the larger or multi-lead ECG datasets by the model which will limit it from adapting to real-world clinical applications. For improvement of this model, these issues need to be addressed i.e. leading it towards end-to-end adaptability and optimizing the architecture.

This study [2] works through combining both ECG and PCG signals in predicting CVD by introducing a multi-modal model using machine learning algorithms. The PhysioNet/CinC 2016 dataset has been used for the research which consists of 409 recordings and data pre-processing is held for balancing the ratio of CVD patients and healthy people through segmenting signals. A deep learning model LSTM networks termed as CL-ECG-Net and CL-PCG-Net are designed for extracting complicated features from both ECG and PCG signals and a genetic algorithm is used for the selection of relevant and significant features. At last, SVM is assigned for feature classification. This model works better than the alternative single-modal model by achieving AUC of 0.936 as the genetic algorithm has shown efficiency in feature selection which leads to better performance. The model has shown relatively poor performance on PCG data the ECG ones as the feature extraction of PCG lacks optimization, not only this the noises from the recordings hampered the data training process. SVM classifiers can be effective but not the best choice for feature classification. As deep learning models follow the black box approaches in case of feature extraction it is confusing to understand which features of ECG and PCG data contribute mostly to the predictive decision. The model's decision is mainly based on the benchmark dataset rather than the real-world data of patients. These drawbacks can be handled through lots of advanced techniques. Wavelet denoising can be used for improving the quality of signals and dynamic segmentation depending on heart rate factors can be used for unbiased portrayal of the activities

of the heart. MFCCs can be used for optimizing CL-PCG-Net to improve its performance.

The paper [3] emphasizes on predicting CVD from datasets that contain a low number of features. The dataset used in this paper is a combination of datasets of Cleveland, Hungarian, Switzerland, Long Beach, Stalag. A sparse autoencoder is used for increasing features of the data. Then 2 different classifiers MLP and 2D CNN are used to predict whether the person has CVD or not. The MLP classifier with SAE gains 89.543% accuracy which is 3.78% higher than MLP without sparse autoencoder. When 2D CNN is used as a classifier with SAE, it achieves 90.09% accuracy. These 2 results are compared to other ML algorithms used to predict CVD such as Random forest, KNN, Decision tree etc and the combination of sparse autoencoder and 2D CNN proves to be the most accurate. Even after combining multiple datasets, it has less than 1000 samples. So it can impact the generalizability of the model. Moreover, it is difficult to interpret how input features influence prediction. So clinical adoption can be hindered because of lack of interpretability.

Hossain et al. [5] introduced a hybrid deep learning framework through the integration of CNN and LSTM networks for capturing spatial and temporal correlations from the data which would lead to CVD detection. Advanced feature engineering and Explainable AI (XAI) are incorporated into this model. Critical factors which would influence the prediction rate are identified using SHAP (SHapley Additive exPlanations). A dataset from Kaggle with 70,000 samples from patients was used where 49.97% were heart disease samples and 50.03% were normal samples. The model achieved an accuracy rate of 74.15% with feature engineering and 73.89% without feature engineering, more than the rate shown by RF, XGBoost and SVM. The study worked on the previous gaps of the former studies by creating a hybrid architecture where CNN and LSTM were used for extracting features and capturing sequential patterns respectively with the help of feature engineering. The model is mainly dependent on a single dataset which is limiting its potential of generalizability over large and diverse datasets. SHAP has been used in the study for in-depth analysis of features but its real application is being missed in case of clinical decision making. Though the accuracy rate is better than some of the previous algorithms, it is not giving the optimal accuracy rate in recent times leaving a lot of scope for improvement. The model needs to be incorporated with some advanced deep learning models as well as semi-supervised learning, it can also be expanded to diverse datasets which would eventually optimize the computational performance improving versatility and scalability of the model.

Lee et al. [6] developed a model which uses multi-modal data for predicting Cardiovascular Disease. Dataset from the Samsung Medical Center (SMC) is used for development and internal validation while UK biobank dataset is used for external validation. 11 different approaches are used and compared to find out the best model with the highest accuracy. DenseNet-169 is trained on fundus photographs (FP) and has an AUROC of 0.548. Logistic regression is trained on only clinical risk factors which attains the AUROC of 0.854. DNN is trained using CRF and gained an AUROC of 0.849. DenseNet-169 and logistic regression is trained on FP and clinical risk factors (CRF) respectively which attains 0.792 AUROC.

DenseNet-169 and DNN are trained on fundus photographs and clinical risk factors which obtains 0.872 AUROC. 10-year ASCVD risk of PCE model was trained on CRF and smoking history which was not present in SMC dataset. It obtains 0.677 AUROC. DenseNet-169 is trained on FP and DNN is trained on FP which attains 0.882 AUROC. DenseNet-169 combined with DNN integrates uncertainty quantification and achieves AUROC of 0.905. Invasive CRF gives better AUROC than non-invasive CRF. However, the study has clear limitations, such as the lack of diversity in datasets, which limit the generalizability of the findings across different populations. Deep learning models consist of multiple hidden layers, due to which the result lacks clinical interpretability and trust. The shortcomings include the lack of consideration for other potentially valuable imaging modalities or risk factors that could further enhance predictions. Future work should focus on incorporating more diverse datasets collected from different regions, developing interpretable models to enhance clinical adoption, and validating these models in real-world clinical settings to assess their feasibility and cost-effectiveness in routine practice.

The study [7] aims at diagnosing heart disease by introducing Heart-Net, a multi-modal deep learning framework which would increase the accuracy rate and consider the relevant factors by integrating the MRI and ECG signals to properly utilize the effect of each method. A 3D U-Net and TCGNN (Temporal Convolutional Graph Network) are used for extracting features from MRI signals and ECG data respectively. The model achieved accuracy rates of 92.56%, 93.45% and 91.89% for the respective 3 datasets HNET-DSI, HNET-DSII and HNET-DSIII outperforming existing frameworks such as fuzzy logic, genetic algorithms etc. which were limited by many constraints such as manual feature engineering. Though the achievements of the model are significant, it has some limitations in adaptability across new datasets. Despite using techniques like ADASYN to reduce imbalanced data, the risk of overfitting prevails due to the complex algorithms of the model. It is quite difficult for the medical professionals to go through the outcomes of the model due to a lack of interpretability. These limitations can be solved by using wearable devices' real time data as well as incorporating transformer-based models that can extract complicated patterns increasing scalability as well as the accuracy rate for detecting CVD in the early stage and other risks of the model. Though the proposed model has shown remarkable outputs, further optimization is required for increasing its robustness and efficiency.

Honi et al. [8] proposed a one-dimensional convolutional neural network approach which uses UCI repository heart disease dataset for predicting CVD at an early stage. After pre-processing the dataset, Chi-square ranking method is utilized to assess the significance of various features in predicting CVD. The model uses convolutional layers for extracting significant features. Then extracted features are fed into a batch normalization layer. For reducing the chance of overfitting, dropout layers are used. The extracted features are flattened and passed through dense layers, with a ReLU activation in the first layer to capture complex patterns and a sigmoid activation in the final layer to output the probability of CVD presence. To evaluate the efficiency of the model, 3 approaches are used. A train-test split, 5-Fold cross-validation and 10-Fold cross-validation. Using the train-test split method, the model gained 99.95% accuracy, 99.93% precision, 99.96% recall and 99.97% F1-score.

5-Fold cross-validation provides less accuracy compared to 10-Fold cross-validation. 5-Fold cross-validation provides 98.53% accuracy rate, 99% precision, 99.92% recall, 99.91% F1-score while 10-Fold cross-validation achieves 99.12% accuracy rate, 99.94% precision, 99.93% recall and 99.98% F1-measurement. The study used the UCI Heart Disease dataset with only 1025 samples, so the model may not have the capacity to generalize to larger, real-world datasets with more complex and diverse patterns. Selecting hyperparameters causes uncertainty. Biased datasets can harm the model’s performance. Using Electronic Health Record (EHR) data raises concerns about data completeness and accuracy, which can impact the model’s reliability. To improve generalization, datasets from multiple sources can be incorporated in future. Also, the impact of the model can be evaluated by testing the model with live patient data.

Liu et al. [9] has shown a proposal of integrating multimodal data with deep learning in diagnosing heart failure. It mainly works by combining multiple factors such as ECG signals, X-rays of the chest and structured text data as it can outperform the single modality models in terms of accuracy rate. A dataset was collected from a hospital which included 440 cases where 325 belonged to heart failure patients and 115 were of mild CVD. The mentioned multimodal model works mainly by integrating features leveraging both feature level and score level fusion. For ECG signal processing, chest X-ray analysis and structured text data, it uses 1D-CNN, DenseNet121 and MLP individually. This model has a high accuracy rate of 99.63% when it was tested with data augmentation and it has also shown robustness against noise which eventually outperformed the former single as well as the dual-modal models, depicting the significance of introducing data fusion in the model. Despite significant outcomes, the model limits some factors like less adaptability to diverse datasets, reducing reliability on the performance due to the absence of external validation and overfitting can be introduced due to extensive use of data augmentation. It also has no clear analysis on handling corrupted data or noises. Future improvement can be done by exploring hybrid fusion techniques that will improve the performance as well as cross data validation needs to be implemented. Adversarial training and expanding the dataset using multi-center collaborations can help to reduce overfitting. GANs (Generative Adversarial Networks) can be used for handling the biases. Lastly, KNN-imputations or Bayesian models can be used to work with corrupted or missing data.

A new multi-modal framework has been introduced in this study [10] which works through combining demographic data with ECG signals for predicting heart disease with an improved accuracy rate. The PTB-XL dataset has been used which contained 21,000 12-lead ECG samples from 18,869 patients. The model is mainly based on working with different signal preprocessing techniques which include IIR (Infinite Impulse Response) filtering and feature engineering integrating with FFT which is needed for extracting pivotal frequency dependent features. Using the MLP (Multi-layer Perceptron) model in the extracted refined features, it showed an optimal performance with precision of 79%, recall of 73% and F1 score of 76% in the real-world clinical scenarios. Traditionally, models like CNN, RNN or any other ensemble model as well as hybrid models like CNN-LSTM have shown high accuracies but they were limited in working with imbalanced data and there were other re-

search gaps which were filled by the mentioned model through integration of refined feature extraction and selection, robust feature engineering and multi-modal techniques. However, the model fails in capturing the complicated patterns fully in the ECG signals as it is more dependent on feature extraction and selection strategies. Moreover, it is not ideal for handling the real-time ECG monitoring system. Future improvements can be made by introducing the model to larger datasets as well as incorporating real-time data from wearable devices for dynamic diagnosis and also integrating MLP with CNN and RNN which would enhance both demographic and ECG data while detecting heart diseases. These future working probabilities can advance the procedure of CVD diagnosis.

The paper [11] proposes a self-attention based transformer model called CardioSentiNet for predicting cardiovascular disease. The dataset used to train the model is generated from a selected group. After preprocessing the dataset, the data is passed to a self-attention based transformation module. Then the transformed dataset is passed to the CNN layer. Then the parameters are fine-tuned. The Training and Optimization step fine-tunes the model's parameters using a training dataset to gain optimal accuracy. Interpretability techniques are added to maintain transparency. Finally, the validation and testing phases assess the model's generalizability and performance on unseen data. Multi-layer neural network with ReLU activation function is used as model architecture. The proposed model is evaluated using R^2 -value and then compared with other models through this evaluation metric. The reported R^2 -value for the proposed model is 0.994 while other models like CardioNet-deep, CNN, RNN etc have less R^2 -value value than the proposed model. The dataset does not represent the diverse population which limits the ability to generalize the result. High R^2 -value raises concerns about overfitting. The study depends on a single dataset without any external validation. In future, broader demographic representation can be incorporated into the dataset To improve dataset, data augmentation method can be used. To guarantee broader applicability, stratified validation can be executed.

Xia et al. [12] proposed ICVD-ACOEDL (Intelligent Cardiovascular Disease Diagnosis) model which uses multiple modalities in diagnosing heart diseases. For feature selection, classification and hyperparameter tuning it uses ACO (Ant Colony Optimization), DLENN (Deep Learning Enhanced Neural Network) and Bayesian Optimization correspondingly. A dataset from Kaggle was used which contained 1190 samples including 629 disease affected and 561 normal samples. The proposed model surpassed the earlier methods like- ANN, SVM, bagging and Naive Bayes with an accuracy rate of 99.71%. ICVD-ACOEDL model has introduced both automated optimization process and feature selection which enhanced the efficiency and scalability of the model in the real-world data. This model will face some challenges like losing important details using ACO though it assists in improving feature selection and using a single dataset will reduce the diversity of the model. Scaling in the case of larger datasets is tough and challenging due to the requirement of detailed hyperparameter tuning in the model. Improvements can be made by integrating methods for identifying outliers and data clustering in order to increase the effectiveness of the classification of the model, exploring transformer-based architectures and expanding datasets for enhanced robustness. By overcoming these issues, the

ICVD-ACOEDL model has the potential to transform CVD diagnostics, providing a scalable and highly accurate approach.

The sole target of this study [13] is to focus on the multi-modal approach by integrating ECG and PCG signals to detect cardiovascular disease rather than using single-modal approaches. This study used the 2016 PhysioNet/CinC Challenge dataset which contained 6 data subsets from training-a to training-f where training-a contained 409 records. These records contained the ECG and PCG signals of both normal people and patients. The DDR-Net model is being introduced in this study which mainly consists of a dual scale feature aggregation module, a deep residual block module and a Bi-LSTM layer. At first, the DDR-Net- Dual Scale Deep Residual Network model had been used for extracting different features from the signals. Dual scale feature aggregation module mainly detects all kinds of patterns small or large and combines them from the signals using Bi-LSTM layer so that all the information can be extracted. Then for eliminating irrelevant and noisy features SVM based RFE (Recursive Feature Elimination) is used and then through cross validation overfitting is reduced. Lastly, after selecting the features through the SVM classifier decision is made whether a heart signal is a normal or abnormal condition. The model showed 91.6% accuracy rate and an AUC of 0.962 which is quite higher accuracy rate than other single-modal and existing multi-modal models. It also showed 92% sensitivity value, 91.1% specificity value and 0.915 F-score combining both ECG and PCG signals. But the failings of this model are that it is mainly specified to a particular dataset and also there is an absence of cross validation which would limit its adaptability to real-world factors. Using SVM-REFCV would cause computational overhead making the model unfitting for larger datasets. Though the model improves the accuracy rate using Bi-LSTM, increasing the complexity and reducing the clarity of the predictions to the professionals. Further improvements can be made by integrating the medical history, demographics and other factors of a patient with the ECG and PCG signals which will increase the accuracy rate of the model.

2.3 Summary of Key Findings

Recent research integrated machine learning and deep learning models with multi-modal data for improving the accuracy of Cardiovascular disease prediction. One of the above-mentioned models uses sparse autoencoders (SAE) to extract features and along with that classifiers like MLP and 2D CNN are used for classification which gained 90.09% accuracy. Multimodal deep learning model that combines fundus images, clinical risk factors, and DenseNet architectures obtains high AUROC (up to 0.905). Transformer-based approaches like CardioSentiNet use self-attention mechanisms and CNNs which has 0.994 R^2 -value. A 1-D CNN model trained on the UCI Heart Disease dataset proved to be 99.95% accurate in predicting whether a person is suffering from CVD or not. By combining ECG and PCG signals DDR-Net was introduced, enhancing the outcome of accuracy rate of 91.6% and AUC of 0.962. CL-ECG-Net and CL-PCG-Net multimodal models used LSTM and genetic algorithms for selecting the features where AUC reached to 0.936. Models like CraftNet handled the imbalanced data of ECG samplings using handcrafted features, showcasing an accuracy rate of 99.63%. Infusing MRI and ECG samples

with TCGNN and 3D U-Net a model named Heart-Net was proposed which attained 93% accuracy. The Ant Colony Optimization and Deep learning combination used by ICVD-ACOEDL achieves a 99.71% accuracy level. Hybrid CNN-LSTM networks utilizing explainable AI methods such as SHAP enhance feature understanding. The combination of demographic information with ECG signals produces a framework which enhances both precision and recall values at 79% and 73%. However, these models also have several drawbacks including lack of interpretability, generalizability due to small datasets, overfitting risks and clinical validation. Moreover, the models simply predict whether cardiovascular disease is present or not. It does not classify the disease into different types. Future improvements should focus on dataset expansion, real-time integration via wearable devices, explainable AI techniques, and optimizing deep learning architectures for scalability and robustness in clinical practice.

Chapter 3

Proposed Methodology

3.1 Design Process or Methodology Overview

The main objective of this study is to create an innovative deep learning model designed to handle the classification of multimodal cardiovascular risks; a process that combines raw electrocardiogram (ECG) signals with structured clinical metadata. The design process for this research adopts a quantitative methodology which is appropriate for the nature of our data (i.e., ECG waveforms and numerical features). The entire research methodology adopted a structured, data-driven approach addressing both technical and clinical demands. The design method was influenced by several critical objectives such as the requirement for precise classification, the integration of diverse data sources, computational performance and the potential for generalization of the model to the real-world clinical contexts.

3.1.1 Problem Definition and Objective Formulation

The first phase involves a precise description of the issue and the establishment of research goals. The primary goal was to design a multimodal deep learning model, one that could take two different forms of input: time-series ECG signals and numeric characteristics such as `qrs_onset`, `qrs_end`, `t_axis`, `p_axis` etc. This dual-input method was driven by the analysis of ECG signals which contain rich temporal patterns showing heart or cardiac function while metadata of patients provide clinical contextual background details. The model had to weigh performance with real-world limits like data availability and access, class imbalance and finite computing resources.

3.1.2 Data Acquisition and Preprocessing

A dataset of ECG recordings along with its clinical metadata was taken out of the MIMIC-IV ECG dataset. Then, signal data were z-score normalized and resampled to the length of 5000 units. Quantitative characteristics, e.g. `qrs_onset`, `p_axis`, `t_axis`, `qrs_end` etc. were also preprocessed by correlation-based feature engineering and imputation and standardized. The derived ECG signal and data in the form of numbers were merged in a single table. These labels were categorized to be one of the three; Normal, Abnormal or Borderline and represented as one-hot vectors.

The dataset was divided into train, validation and test subsets through stratified sampling to maintain the distribution of classes.

3.1.3 Model Architecture Design

The model architecture was a dual-input neural network with two specialized branches. One branch designed for time-series ECG signals, used a mix of one-dimensional convolutional layers and blocks of dense layers. Convolutional layers help in spotting local features and patterns in the signal; meanwhile, dense layers learn patterns and make decisions based on the input they receive. Also, there are Batch Normalization, Dropout, ReLU activation and Residual Connections for stability plus generalization. Additionally, there is Global Average Pooling for dimensionality reduction and fusion preparation. This set-up helped the model to learn spatial plus temporal features within ECG data quite well. The other branch was a multi-layer perceptron (MLP) which handled the numerical metadata. It had fully connected layers with ReLU activation functions, batch normalization and dropout for regularization. After their outputs had been computed and processed in their respective branches, the outputs of both branches were concatenated into a single feature vector at the fusion layer. This combined representation was then passed through more dense layers just before the final classification layer that employed a softmax activation function to generate class probabilities. In this way, the architectural design enabled the model to learn features from both modalities that are complementary and thus make wiser predictions.

3.1.4 Model Training Strategy

The model training process with the dataset was tuned properly and carefully in order to enhance performance and avoid overfitting. Based on validation loss, Early Stopping was implemented to prevent unnecessary or irrelevant training. Class Weights were implemented which were automatically calculated using the `compute_class_weight` function from Scikit-learn in order to balance the imbalanced dataset or class and applied during training process. In this way, underrepresented classes made more contributions significantly to the loss function and hence improved sensitivity by the model to minority classes. Training throughput was further enhanced by mixed precision that enabled faster computation and lower memory consumption through 16-bit floating-point operations wherever possible. Iterative adjustments were made to batch size, model depth and dropout rates to achieve optimum performance while fitting within GPU memory limits.

3.1.5 Model Evaluation and Interpretation

After training the model with the dataset, the evaluation of the model was determined by using standard classification metrics which includes accuracy, precision, recall and macro-averaged F1-score. Based on validation accuracy, the best model or architecture was saved in a lightweight .h5 format for better performance at the time of testing. Confusion matrix was generated at the end of the evaluation in order to analyze the distribution of true and predicted labels which ultimately provides insight into the model's strengths and limitations across different classes.

In summary, the design process adopted here makes certain that the model developed is accurate, interpretable and practically feasible when applied to real-world cardiovascular risk assessment. This process takes a stride majorly toward automated, reliable and personalized cardiac diagnostics by fusing multi-modal data via a carefully engineered deep learning framework.

3.2 Preliminary Design or Design (Model) Specification

In this section, we have explored several deep learning frameworks designed to carry out multi-modal evaluation of cardiovascular risk from both ECG signals and numerical clinical data. The goal is to survey various signal-processing models that can capture spatial and temporal patterns in the ECG waveform. The models are being evaluated and compared based on their performance in an ECG classification task.

Each design centers on handling the raw ECG signals of set length which is 5000 samples as the related numerical (tabular) data is managed apart via a unified MLP (Multilayer Perceptron) pipeline discussed at the end of this section. The four signal frameworks studied are CNN + LSTM, DenseNet, MobileNet and WaveNet. They were chosen based on their well-documented results in time-series and biomedical signal-processing assignments.

3.2.1 CNN + LSTM

CNN + LSTM model is a hybrid deep learning framework which merges **Convolutional Neural Network (CNN)** with **Long Short-Term Memory (LSTM)** network. CNN captures the local spatial features efficiently in the time-series data—for example, peaks, slopes and intervals of an ECG signal. It applies convolutional filters over short temporal windows whereas LSTMs are designed to model long-range dependencies in sequential data keeping internal memory across time steps.

In this model architecture, the ECG signals initially go through multiple 1D-convolutional layers that extracts nearby waveform characteristics. These intermediate representations are then used in LSTM layers which learn how these features change with time. Because of this combination, the model is able to identify both long-term rhythm patterns and short-term morphology in the ECG signal.

The CNN + LSTM model is especially appropriate for biological data where both spatial structure and temporal growth are clinically relevant. But, it demands more time for training and greater computation because of the sequential nature of LSTMs.

3.2.2 MobileNet

MobileNet is a simple convolutional neural network design suited for resource-limited environments. It attains great efficiency by using depthwise separable con-

volution which greatly cut down the number of parameters and operations when compared with standard convolutions.

In the context of classifying ECG signals, MobileNet is adopted in its 1D version. It processes signals at a very high speed with minimal computational overhead; preserving the essential features required for classification. Owing to its compact size, MobileNet becomes very handy when working with smaller datasets or deploying models in real-time/edge computing environments.

MobileNet’s speed and efficiency made it an attractive rival for rapid prototyping and low-latency applications.

3.2.3 DenseNet

Densely Connected Convolutional Network (DenseNet) presents a novel architecture in which layers receive connections from all preceding layers in a feed-forward fashion. Such a densely connected design facilitates the reuse of features and reinforces the gradients’ flow which allows even very deep networks to be constructed efficiently.

For ECG classification, a 1D variant of DenseNet is used to handle the unprocessed signal. The design allows the model to learn tiered representations capturing both fine details and overarching abstract characteristics of ECG waveforms. Dense connections minimize redundancy and increase learning efficiency while avoiding vanishing gradients.

DenseNet’s value comes from its capacity to produce abundant and varied features which enhances classification quality. However, this results in greater memory utilization and slightly longer training times.

3.2.4 WaveNet

WaveNet is a deep convolutional model first developed for the synthesis of raw audio. It uses dilated causal convolutions to simulate long-range temporal connections in time-series data without recurrent layers. The dilation increases exponentially the receptive field with depth because of which the model can observe a wide time span of the signal.

For ECG signals, WaveNet applies stacks of 1-D convolutional layers with dilation rate that process the entire signal in parallel. This parallelism significantly increases the speed of training on GPUs. WaveNet can capture temporal dynamics while being computationally efficient unlike RNN-based models.

WaveNet showed great results in ECG classification. It gives an optimum balance between temporal modeling and computational cost. The way it is built lets it generalize well over long signals without having to wait because of recurrent operations.

3.2.5 Multilayer Perceptron (MLP)

Multilayer Perceptron (MLP) is a type of feed-forward artificial neural network which is commonly used for tasks like supervised learning. MLP generally contains an input layer, one or more hidden layers with nonlinear activation functions and an output layer. The neurons in a layer connect completely to the neurons in the next layer. In this way, the network can learn complicated mappings from inputs to outputs.

In all the proposed designs, the numerical clinical features (like qrs_onset, t_axis, p_axis etc.) go through a Multilayer Perceptron (MLP). This part has fully connected dense layers with ReLU activations and dropout regularization. The MLP serves as a parallel input that learns representations from the tabular clinical data.

After processing the ECG signal by its respective deep learning model (CNN + LSTM, DenseNet, MobileNet, WaveNet), extracted features are fused with the output of the MLP branch. This fusion is typically done through concatenation and then additional dense layers to perform the final classification.

The use of shared MLP across all architectures ensures uniform handling of tabular data making fair comparison among the models possible and allowing the joint learning of waveform and clinical patterns resulting in more accurate cardiovascular risk assessment.

3.2.6 Simulation and Verification

Simulations were conducted through training and evaluating the models on the ECG dataset in order to verify the performance and effectiveness of each proposed model. Each model architecture was trained properly using stratified data splits and evaluation are performed based on accuracy, precision, recall and F1 score. These metrics provide a complete picture of categorization performance. The simulation results validated each model's respective strengths and limitations which influenced the ultimate model decision.

Characteristic	CNN + LSTM + MLP	DenseNet + MLP	MobileNet + MLP	WaveNet + MLP
Description	Combines spatial and temporal learning	Deep hierarchical features	Lightweight model	Models long-term dependencies
Accuracy	Moderate	High	Good	High
Efficiency	Slower	Moderate	Very High	High
Robustness	High	High	Moderate	High
Suitability	Strong temporal learning	Best for large datasets	Ideal for real-time setups	Excellent for long signals

Table 3.1: Performance Comparison of Models

3.3 Data Collection

For this research purpose, we used Physionet MIMIC-IV ECG dataset [4]. Initially, the MIMIC-IV-ECG module comprises around 800,000 diagnostic electrocardiograms (ECGs), recorded from approximately 160,000 distinct patients. Each ECG is a 10-second recording captured using the standard 12-lead configuration and sampled at a frequency of 500 Hz. Diagnostic ECGs are recorded at Beth Israel Deaconess Medical Center (BIDMC) as part of regular care. In our dataset, 3 types of information are present.

- (i) The electrocardiogram waveforms themselves
- (ii) the machine measurements (ex: average RR interval as calculated by the machine)
- (iii) the cardiologist reports

For each diagnostic ECG, the machine produces a summary report along with key measurements such as the RR interval, QRS onset etc. These outputs are collectively referred to as machine measurements. The outputs generated by the machine are parsed to remove protected health information (PHI). The machine measurements for each ECG waveform are provided in the `machine_measurements` file. There is a data dictionary which provides the description for each of the initial features in `machine_measurements_data_dictionary`. The `machine_measurements` file provides the machine generated reports in columns `report_0` to `report_17`. Each row of `record_list` file has a path which directs towards the corresponding `.hea` and `.dat` files.

The quality and accuracy of the MIMIC-IV-ECG dataset are ensured through multiple measures. The ECGs are recorded in a clinical setting using standardized 12-lead machines at a consistent sampling rate (500 Hz) which maintains signal quality. Machine-generated measurements follow validated algorithms widely used in clinical practice while cardiologist reports offer expert validation. The inclusion of detailed data dictionaries and structured CSV files supports consistency, traceability and reliability for research use.

After extracting an ECG signal using WFDB package, we get the following:

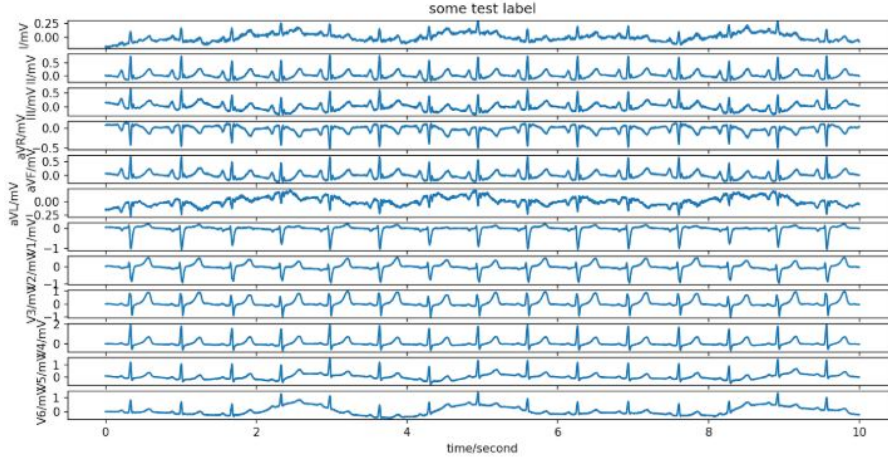


Figure 3.1: A 12-lead ECG Signal

3.3.1 Data Cleaning

The tabular dataset didn't have any null rows in relevant features. But there were more than 0.1 million null rows in `ecg_label`. Those rows were removed from the dataset. Some outliers were present in the numeric features which contained an abnormal value (29999). After deleting the rows with outliers in `p_axis`, we found that none of the numeric features had any outliers present anymore.

After loading the raw ECG signal data, we convert it to `float32`. ECG signals are extracted as a numpy array from the waveform file. If the ECG signal has too many (5% of the signal) missing (`NaN`) values, the signal is discarded. But if it contains a comparatively less number of `NaN` values, then the `NaN` elements are replaced with 0.

3.3.2 Signal padding or truncating

Most deep learning architectures and other sequence-based models require inputs of same shape to perform efficient batch processing and enable parallel computation on hardware accelerators such as GPUs. Without equal input shapes, the model would face difficulties during training. Because variable-length signals would disrupt matrix operations and prevent the formation of coherent mini-batches. If the signal is too long or too short, the signal is truncated or padded respectively to 5000 length. The threshold of 5000 was chosen so that we do not lose much information from an ECG Signal. The short signals are padded with extra 0 to increase the length.

3.3.3 Data Transformation

Standard scaling transforms the feature in a way that the mean of that feature is 0 and the standard deviation is 1. In the tabular dataset, different numeric features have different ranges of values. If not scaled properly, the features with larger value

can dominate the learning process and features with smaller values might be ignored by the model. Since the numeric features had gaussian distribution with less outliers, standard scaling is performed to transform the values of the numeric features.

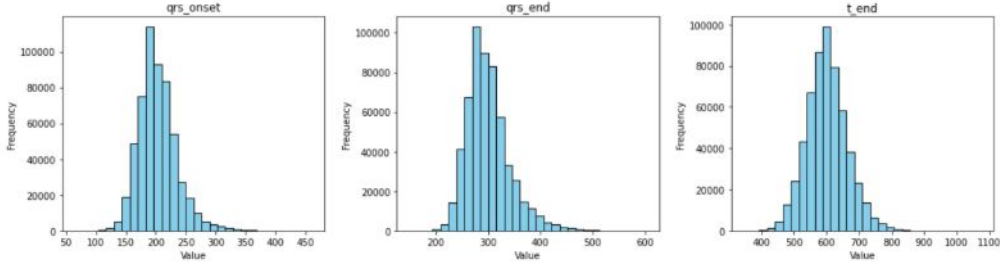


Figure 3.2: Histogram of Different Numeric Features

For a given feature x ,

$$z = \frac{x - \mu}{\sigma}$$

where,

x = original numeric value of the feature

μ = mean of all the values in the feature

σ = standard deviation of the feature

z = scaled value of the feature

We performed z-score normalization on the ecg signals per lead from the waveform file so that their value falls within a standard range. ECG leads might have different amplitude ranges. Normalization prevents the leads with larger value to dominate the training of the model.

3.3.4 Feature Engineering

To enhance the model's ability to differentiate between Normal, Borderline and Abnormal ECGs, several new features were engineered from the original ECG machine measurements. The new features are: qrs_duration, pr_interval, qt_interval, p_axis_deviation, qrs_axis_deviation, t_axis_deviation, qrs_duration refers to the time required for the ventricles to depolarize. When qrs_duration is less than 120 ms, it is in normal range. It indicates that the signal is passing through the ventricles at a healthy speed. If the signal is too long then it means the signal is being slowed or blocked due to a heart problem.

pr_interval is a vital ECG measurement that indicates the time duration from the start of the p wave to the start of the qrs complex. It shows how much time is required for the electrical signal to pass from the atria through the AV node into the ventricles. If the duration is too long, it may be a sign of heart block. If it is too short then the heart will not have enough time to be filled with blood before pumping. Therefore, it is a significant indicator of normal or abnormal ECG labels. qt_interval represents the time it takes for the ventricles to squeeze and then relax. If the qt_interval is too long or too short, it indicates irregular heartbeat (Arrhythmia). Therefore, it was also added as a feature.

p_axis is the angular direction of the electrical signal while the signal passes through the atria during atrial depolarization. The normal range of the p_axis is between

0 degree and around 60 degree. If the signal is travelling through the atria in an unusual direction, then it will deviate from the usual range. It indicates how much the p_axis deviates from the normal. Therefore, it was added as a numeric feature. Similarly qrs_axis_deviation shows the unusual direction for ventricles and t_axis deviation represents the deviation from usual angular direction when ventricles relax. Therefore, they were added in our dataset to enhance the ability to classify different types of ECG labels.

3.3.5 Data Reduction

In the tabular dataset, there were some columns which were irrelevant to our research. For example, cart_id, ecg.time, bandwidth and filtering. We dropped these irrelevant columns. ECG labels were found in the columns of generated reports. We extracted the labels from those columns and created a new column called “ecg_label”. Then we dropped the irrelevant columns containing generated reports. Record_lists and machine_measurements were merged based on subject_id and study_id so that the path to waveform file is added in the tabular dataset. Then subject_id and study_id are dropped to reduce redundancy. Based on the correlation matrix, it was found that rr_interval, p_onset, p_end, p_axis has negligible relation with ecg_label because the correlation between these numerical features and ecg_label is less than 0.1. We selected 0.1 as a threshold for choosing relevant features. So, the irrelevant columns were dropped.

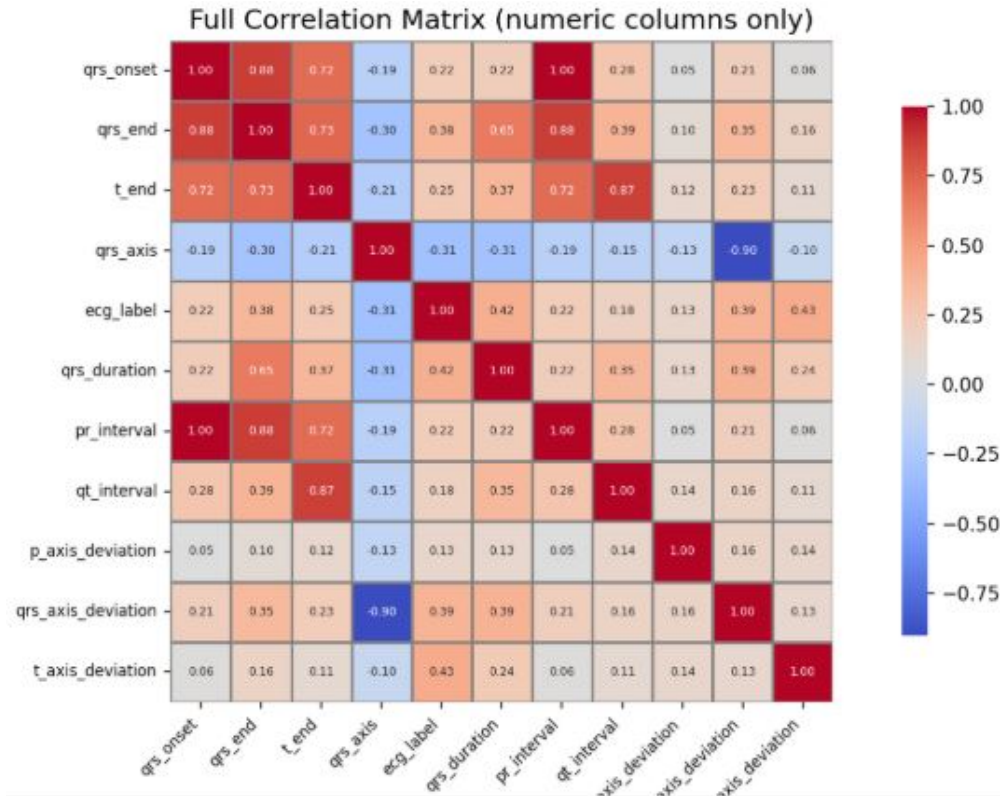


Figure 3.3: Correlation Matrix

3.3.6 Encoding

Label encoding was used to map the categorical ECG labels to an integer value. Normal ECG, Borderline ECG and Abnormal ECG are represented as 0, 1 and 2 respectively. Then, one-hot encoding was implemented to convert those integers to one-hot vectors. This ensures numerical stability and properly formatted target variables for multi-class classification.

3.3.7 Summary of Preprocessed Data

The final pre-processed tabular dataset contains 564761 rows with each row containing a path directed towards the ECG signal waveform and the numeric features qrs_onset, qrs_end, t_end, qrs_axis, qrs_duration, pr_interval, qt_interval, p_axis_deviation, qrs_axis_deviation and t_axis deviation. Target variable is the ecg_label column.

After pre-processing:

qrs_onset	qrs_end	t_end	qrs_axis	ecg_label	path	qrs_duration	pr_interval	qt_interval	p_axis_deviation	qrs_axis_deviation	t_axis_deviation
-1.022382	0.988528	-1.326337	1.50378802	1	files/p1000/p10000032/s40689238/4068923	-0.422832887	-1.021797495	-1.10157359	-0.019146335	-0.029538906	0.013143
-1.271029	-1.28796	-1.550502	1.45522437	1	files/p1000/p10000032/s44458630/4445863	-0.621690059	-1.27025722	-1.23391858	-0.184000332	-0.586320361	0.257784
-1.271029	-1.336198	-2.030854	1.38237889	0	files/p1000/p10000032/s49036311/4903631	-0.721118645	-1.27025722	-1.8956435	-0.101573334	-1.071492544	-0.067508
-0.711575	-1.095005	-1.006103	1.23668794	0	files/p1000/p10000117/s45090959/4509095	-1.11883299	-0.71122284	-0.88099862	-0.101573334	-1.241836909	-0.284066
-0.151212	-0.419666	-0.045399	1.57663350	0	files/p1000/p10000560/s41445586/4144558	-0.621690059	-0.152188461	0.0454162735	0.599056153579	-0.844366723	-0.257784
-0.462928	-0.516143	-0.109446	1.72232445	1	files/p1000/p10000560/s42695383/4269538	-0.323404301	-0.462763116	0.1777612585	0.516629155086	-0.674022358	-0.067508
-0.214282	-0.130235	0.14674170	-0.778704	1	files/p1000/p10000635/s40067704/4006770	0.07431004295	-0.214303392	0.3542212386	-0.84341632	0.773904747944	0.20431556525504
-0.773736	-0.61262	-0.269563	-0.633013	2	files/p1000/p10000635/s45386375/4538637	-0.025118543	-0.773337771	0.1777612585	1.258472141529	0.603560382564	-0.230601
-0.960221	-0.757336	-0.37090591	-0.754422	2	files/p1000/p10000635/s43522917/4352291	-0.025118543	-0.959882564	1.1924061405	-0.678562323	0.745514020380	0.23149787199722
-0.640413	-0.467905	-0.27483553	-0.584449	1	files/p1000/p10000635/s42947358/4294735	0.07431004295	-0.649107909	0.8394861816	-0.554921825	0.546778927437	0.39459171245031
-0.400767	-0.27495	-0.94732818	-0.73014	2	files/p1000/p10000635/s48339811/4833981	0.07431004295	-0.400648185	1.5894410943	1.01191146048	0.717123292817	0.53050324616122
1.21543409	0.68982034	0.08269476	0.77533326	2	files/p1000/p10000764/s47218930/4721893	-0.522261473	1.21434002250	-0.74865363	-0.142786833	-1.043101816	-0.475242
0.96678781	0.30391206	-0.23754	0.50823318	2	files/p1000/p10000764/s44069449/4406944	-0.919975817	0.96588029830	-1.0133436	-0.019146335	-0.730803813	0.47613863267686

Figure 3.4: Preprocessed New File

Besides the waveform path and measurements, the data set got cleaned and standardized so that it would be consistent across all the records. Missing or strange values were dealt with in a proper way so that data integrity would be maintained. This data set acts as the input for training and testing multi-modal deep learning models which combine both signal and clinical feature modalities for ECG classification tasks.

3.4 Implementation of Selected Design

The implementation phase converts the proposed system design into a solution that is functional and operational. This process has three major stages: development, training, and testing. In the final architecture, two-input models handled raw ECG signals with deep learning models like CNN+LSTM, DenseNet, MobileNet and WaveNet; meanwhile, it handled tabular numerical features with Multilayer Perceptron (MLP).

3.4.1 Numerical Branch

Multilayer Perceptron (MLP)

In the numerical branch, normalized features are passed to a Multilayer Perceptron (MLP). The first layer is a dense layer with 128 units with ReLU activation and BatchNormalization followed by Dropout(0.3) - currently representing more concretely the available mechanism which randomly disables 30% neurons in training to

reduce overfitting and hence improve generalization. The second dense layer has 64 units also with ReLU activation and BatchNormalization followed by Dropout(0.3) again. More conventionally, 128 and 64 units represent a typical compression of the feature space whereby the network gets to learn increasingly more abstracted representations of the input data.

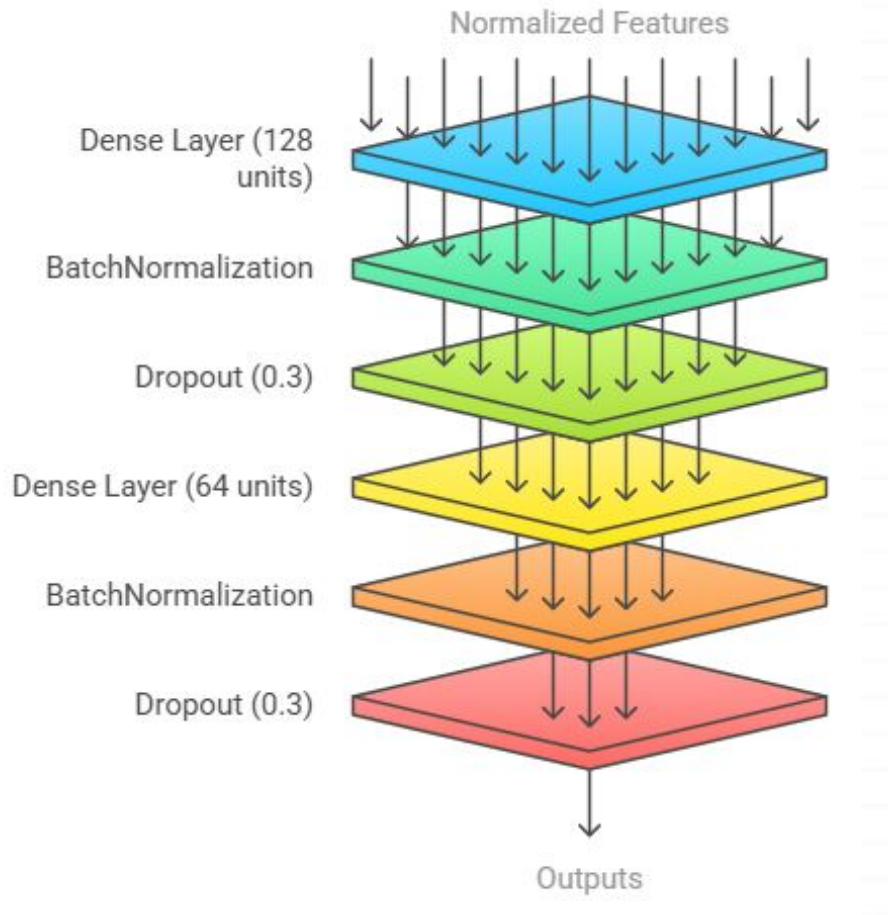


Figure 3.5: Multilayer Perceptron (MLP)

3.4.2 Numerical Branch

CNN + LSTM

1. CNN+LSTM architecture with MLP is a two-input deep-learning model that utilizes ECG signals along with clinical numerical data processed at once.
2. The ECG branch entails the processing of 12-lead sequences using a well-formulated cascade of 1D-convolutional layers and long short-term memory (LSTM) layers.
3. One Conv1D layer with 128 filters, the kernel size of 5 and a rectified linear unit (ReLU) activation, attempts to capture local analysis like P waves, QRS complexes and T waves.
4. The next MaxPooling1D layer containing 2-pool size is used to reduce dimensionality and also keeps mostly salient features.

5. Thereafter, a second Conv1D layer with 256 filters is added to derive more abstract representations which is again followed by MaxPooling1D having a pool size of 2.
6. The advantage of a third layer of Conv1D (128 filters) is in the possibility to further define the extraction of patterns and another MaxPooling1D (2) layer further reduces the dimensionality.
7. The volume of the output is ultimately directed to an LSTM layer that has a capacity of 128 units, that is clearly modeling the long-range timelines present in the ECG time-signals; the temporal sequence of the waveforms may include vital information that makes it a critical indicator of a diagnosis.
8. Before the fusion of the input with the MLP branch, the LSTM output is processed in a Dense layer with 64 units and a Dropout layer that generates a dropout rate of 0.3 with the clear goal of avoiding the overfitting phenomena.

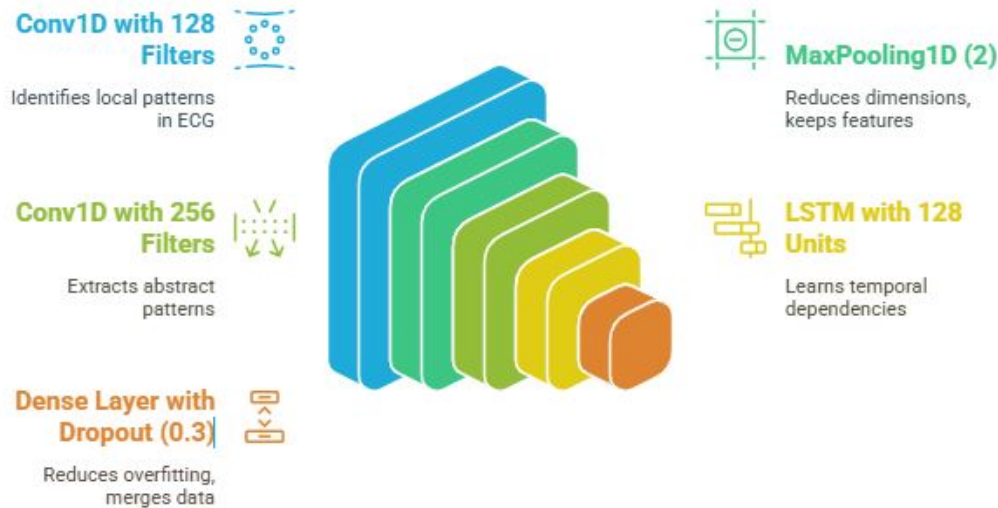


Figure 3.6: CNN + LSTM

MobileNet

1. The current model uses a 1D MobileNet-like structure that is modified to work with 1D data (electrocardiographic (ECG) signals) as opposed to 2D images.
2. The input ECG is possession of 12-lead reads that are compounded into a number of 5000 time spans in each lead.
3. The first layer (Conv1D) has 128 filters and kernel size of 3 to reduce the temporal resolution so as to conserve important local patterns.
4. Later BatchNormalization and ReLU activation jailbreak learning and add the non-linearity.

5. The next signal chain consists of blocks of MobileNet where each block takes the form DepthwiseConv1D (a 1-dimensional depthwise convolution) and is combined with pointwise Conv1D (pointwise convolution 1x1).
6. The size of the parameters is drastically reduced in a Depthwise separable convolution although the model still has the capacity of learning sophisticated patterns.
7. DepthwiseConv1D treats individual input channels separately and in so doing, learns channel-wise spatial features but the pointwise Conv1D incorporates cross-channel interactions.
8. Both convolutions then have BatchNormalization and ReLU activation which make it easier to achieve training and stable activation distributions.
9. The multiplication of the filters in the different layers (grows as 64 to 512) enables the network to learn the features in different levels of abstraction.
10. Stride 2 Temporal downsampling expands receptive fields and learns longer range dependencies.
11. The last signal branch is “GlobalAveragePooling1D” and reduces its temporal information down to one feature vector.
12. This vector is fed into a 128-units Dense layer with ReLU activation to come up with a high-level representation.
13. To prevent the over-fitting; a Dropout layer with probability of 0.3 is added after the Dense layer which gives strong regularization especially in the deep and wide signal extension.

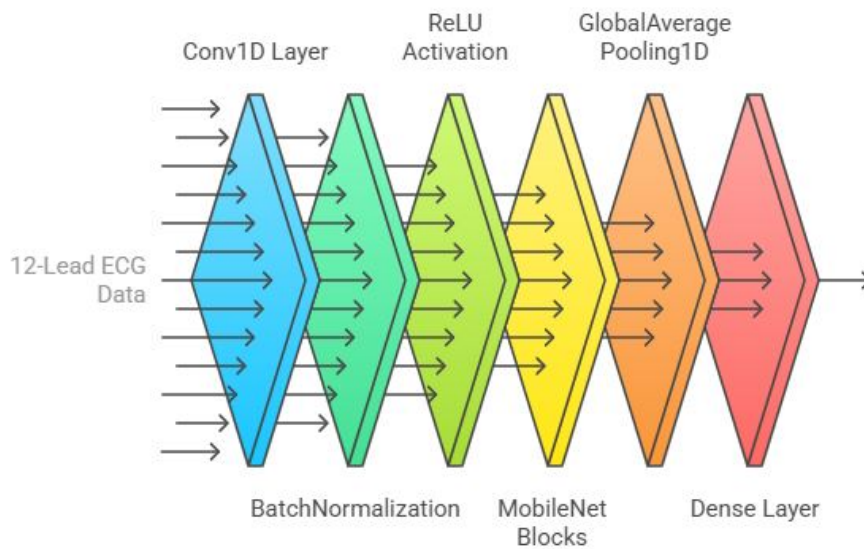


Figure 3.7: MobileNet

DenseNet

1. The ECG signal branch adopts the DenseNet-like structure that is designed to favor not only feature reuse but also gradient propagation.
2. The first raw signal is fed into a Conv1D layer of 64 filters and a kernel of 7 which allows the extraction of low level temporal features on all the 12 leads.
3. The successive BatchNormalization is used to stabilize and accelerate the learning and the ReLU activation which introduces the non-linearity.
4. The 1D MaxPooling layer with a pool size of 2 is added to reduce the time dimension yet keeping the significant data. The signal is then passed onto three thick blocks which have four layers which are sequential.
5. In a dense block each of its layers is composed of BatchNormalization, ReLU activation and a Conv1D layer, 32 filters and a kernel size of 3.
6. The last layer output of any given dense block is concatenated to outputs of all the previous dense blocks in the same dense block therefore promoting feature reuse.
7. Following every dense layer, a transition layer is added that will squeeze feature maps and limit overfitting.
8. The 1x1 Conv1D followed by the AveragePooling1D with the pool size of 2 is present in each transition layer.
9. Compression is used in the form of a factor of two (i.e., 0.5) which indicates that the number of filters is reduced by half in order to manage model size.
10. After the three dense blocks there is a GlobalAveragePooling1D layer which summarizes the features on all resolved temporal steps into an individual vector.
11. The resultant representation is transferred to another Dense layer of 128 units and ReLU activation.
12. A dropout layer containing rate of 0.3 is inserted for the reduction of overfitting before combining with the MLP branch.

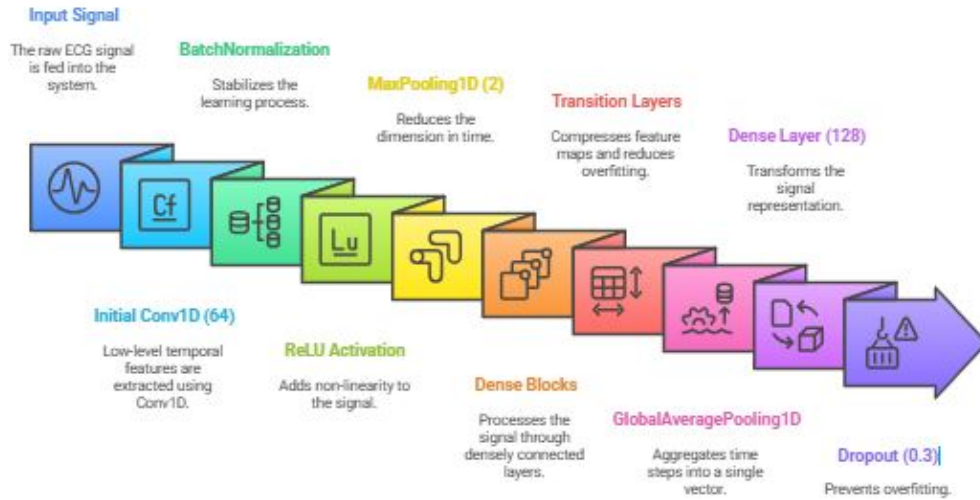


Figure 3.8: DenseNet

WaveNet

1. WaveNet-like architecture has also been developed in the field of electrocardiogram (ECG) signal-processing and adapted to support 1D ECG signals.
2. This component is fed by the 5000-sample data collected in 12 leads.
3. First convolutional skill uses 128 filters and a 2 size kernel and ReLU activation to extract low-level features of the raw input.
4. The next stage is a back-and-forth dilated convolution with 1, 2, 4, 8, 16, 32, 64, 128, 256 and 512 dilation factors.
5. There will be a gated activation unit per level of the dilated layer to control the passage of information.
6. The result is divided into two branches; a last convolutional layer matching the doubled residual channels parameter is used to fit the widened channel dimension.
7. The Filter component is constructed by tanh activation of the first branch whereas the second branch is activated by sigmoid and it becomes the Gate.
8. A gated activation is obtained by multiplying element-wise between the Filter and Gate.
9. Every gated unit will yield two results:
10. A residual term applied in the form of a 1 x 1 convolution and intensified to the input ($x = x + \text{residual}$).
11. A skip term, applied as a 1 1 convolving and saved so as to be combined later.
12. The skip connections ensure that information can be regarded in numerous receptive fields and then re-compose, allowing local and long-range dependence-based learning to be learned.

13. After processing all the dilations the skip terms are added with the `tf.add_n()`.
14. This is then collapsed into a single direction further into a feature vector with the use of a `GlobalAveragePooling1D` layer.
15. The generated representation is passed through a Dense layer having 64 units with ReLU activation.
16. Then, a Dropout layer that has the value of 0.3 is added to avoid overfitting.

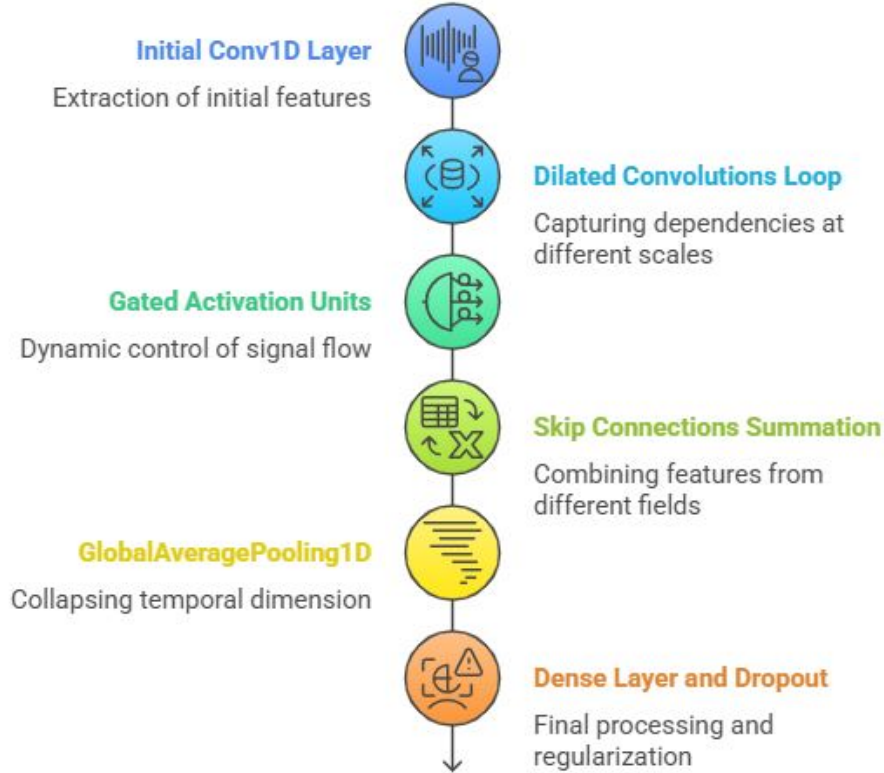


Figure 3.9: WaveNet

3.4.3 Concatenation

The products of the two branches are concatenated to a combined feature representation and preferably passed into a Dense layer of 128 units and ReLU activation with a Dropout (0.3) layer. This enables the model to take an account of the connection between the signal patterns and clinical metadata. The last layer is the output layer, a Dense (3) layer with softmax activation that reformats the learned features into a probability distribution in the three target classes as Normal, Borderline or Abnormal. In this case, it is reasonable to use softmax since it allows the results to be represented as the probability of classes in a multi-class classification task.

3.4.4 Training

The float32 and float16 are used together as mixed precision to accelerate computations with compatible GPUs to enhance the streamline training process. Training

data pipeline is used utilising `tf.data`. Dataset to support loading and preprocessing. Each signal is loaded using a `load_signal()` function as a `tf.py` function to load, pad/trim and normalize the ECG signal by z-score. The data has been stratified and divided into three parts 80 percent training, 10 percent validation and 10 percent testing with equal distribution of classes within each subset. The data has been shuffled then it has been batched in size 128 and prefetched to enhance GPU utilization and throughput.

3.4.5 Model Compilation

Model optimization was carried out using Adam optimizer since it has an adaptive learning rate. Loss function is `categorical_crossentropy` which fits multi-class problems and accuracy is being used to monitor performance. Due to the class imbalance problem and especially regarding the underrepresented 'Borderline' class, automatic class weighting was applied. The weights were computed by calling `compute_class_weight` ('balanced') from `scikit-learn` which in turn adjusted the contributions of losses during training so that all classes were weighted fairly.

Three major callbacks were included to handle the training. `EarlyStopping` served to keep track of `val_loss` and terminated the learning when no progress occurred 8 times in a row, overfitting was reduced and compute time was saved. The best model was stored using the `ModelCheckpoint` method taking into account the `val_accuracy`; thus, the most effective one was kept. `ReduceLROnPlateau` tracked `val_loss` and reduced the learning rate by 0.2 when it had plateaued 3 epochs which would make smaller steps toward convergence with a more accurate change without missing local minimums. Training of the model was carried out up to 35 epochs but usually led to early stopping at many fewer epochs.

3.4.6 Testing

The best-performing model after training was loaded with `load_model()` and tested on the test set. Metrics for evaluation included loss and accuracy besides a classification report from `sklearn's classification_report()` that details each class's precision, recall and F1-score. This elaborates how well the model ranked across classes — a much fairer evaluation than accuracy alone. A confusion matrix was also computed for visual misclassification inspection — to help track systematic errors like confusion between 'Borderline' and other classes.

3.4.7 Evaluation Metrics

DenseNet + MLP

Test Accuracy: 0.8804

Characteristic	Normal	Borderline	Abnormal	Macro Avg	Weighted Avg
Precision	0.85	0.81	0.95	0.87	0.88
Recall	0.91	0.83	0.93	0.89	0.89
F1-Score	0.88	0.82	0.94	0.88	0.89
Support	15421	17067	23989	56477	56477

Table 3.2: DenseNet + MLP Evaluation Metrics

The confusion matrix is as follows:

$$\begin{bmatrix} 13922 & 1334 & 165 \\ 2098 & 13909 & 1060 \\ 327 & 1768 & 21894 \end{bmatrix}$$

WaveNet + MLP

Test Accuracy: 0.8711

Characteristic	Normal	Borderline	Abnormal	Macro Avg	Weighted Avg
Precision	0.85	0.81	0.93	0.86	0.87
Recall	0.89	0.82	0.91	0.87	0.88
F1-Score	0.87	0.81	0.92	0.87	0.87
Support	15421	17067	23999	56477	56477

Table 3.3: WaveNet + MLP Evaluation Metrics

The confusion matrix is as follows:

$$\begin{bmatrix} 13778 & 1393 & 250 \\ 1852 & 13656 & 1559 \\ 316 & 1910 & 21763 \end{bmatrix}$$

CNN + LSTM + MLP

Test Accuracy: 0.8528

Characteristic	Normal	Borderline	Abnormal	Macro Avg	Weighted Avg
Precision	0.83	0.77	0.94	0.85	0.86
Recall	0.89	0.80	0.87	0.85	0.85
F1-Score	0.86	0.78	0.90	0.85	0.85
Support	15421	17067	23989	56477	56477

Table 3.4: CNN + LSTM + MLP Evaluation Metrics

The confusion matrix is as follows:

$$\begin{bmatrix} 13756 & 1503 & 162 \\ 2347 & 13610 & 1110 \\ 526 & 2663 & 20800 \end{bmatrix}$$

MobileNet + MLP

Test Accuracy: 0.8491

Characteristic	Normal	Borderline	Abnormal	Macro Avg	Weighted Avg
Precision	0.84	0.77	0.94	0.85	0.86
Recall	0.88	0.79	0.88	0.85	0.85
F1-Score	0.86	0.78	0.91	0.85	0.86
Support	15421	17067	23989	56477	56477

Table 3.5: MobileNet + MLP Evaluation Metrics

The confusion matrix is as follows:

$$\begin{bmatrix} 13579 & 1693 & 149 \\ 2003 & 13502 & 1562 \\ 621 & 2494 & 20874 \end{bmatrix}$$

Chapter 4

Conclusion

Advanced technology cannot entirely be sufficient to fight against Cardiovascular diseases (CVDs) because they demand significant transformation from the prevailing diagnostic and treatment methods. This research tries to establish new methodologies and protocols for detecting CVD in an early stage extending the horizon of technical discoveries and exploring Deep learning methods in a multi-modal framework. The current research discussed deep-learning based ECG classification practices including raw 1D ECGs and numerically generated variables as suggested by clinicians. It used a dual-input framework in which signal characteristics are incorporated together with structured predictors into a common framework. There were five different signaling models tested, a CNN-LSTM composite, a DenseNet-style encoder, a WaveNet-like sequence module and a version based on MobileNet. All these designs were built in such a way that they were able to host their temporal-based feature extractions as well as inductions of long-range temporal dependence. The numerical data were evaluated parallel using a multilayer perceptron and allowed the simultaneous usage of signal and structured information. The notable advantage of the methodology is in handling intricate patterns in large datasets. The model distinguishes the patterns that can be overlooked by the single modal method because it evaluates multiple informational sources at once. Yet, the journey is far from over. The predictive model requires extensive computational resources because it currently performs better with populations that match training data sources. Research moving forward requires improvements making the model operate with real-time device data while making the system scalable enough for broad implementation.

In summary, this research project goes beyond technical contribution because it leads to a tomorrow where Cardiovascular diseases will become preventable and better understood through anticipation. This research plans to redesign cardiovascular medical care by connecting AI systems with clinical expertise in order to better cardiac treatments.

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